Ink Legends

A plain-English guide to changing the game yourself, and getting it onto your phone.

Version v0.2 **Godot 4.7** **No coding needed**

1. Before you change anything
2. How to open and edit the files
3. Every fighter and their files
4. Damage of basic attacks
5. Damage of specials
6. Character abilities and passives
7. Health
8. Coins (gold) and rewards
9. Unlocking fighters
10. Rarity tiers and chests
11. Maps, wind and gravity
12. Names, lines and patch notes
13. Testing your change
14. When something breaks
15. Putting the game on Android
16. iPhone, and playing in a browser

The game folder used in this guide is `D:\gam`. If you move it, every path below starts from your new folder instead.

1. Before you change anything

> **Make a backup first.**
> Copy the whole `D:\gam` folder somewhere safe (for example `D:\gam_backup`). If anything goes wrong, copy it back.

Almost everything you will want to change is a **number** in a text file. The golden rules:

1. **Only change the number.** Keep everything around it exactly as it was: the name, the `=`, the `:=`, quotes, commas and brackets.
2. **Keep decimals as decimals.** If it says `470.0`, write `500.0`, not `500`. If it says `30`, write a whole number like `25`.
3. **Never change the tabs at the start of a line.** The game reads the indentation, and a missing tab breaks the file.
4. **Close the game before editing your save file.** The game rewrites the save when it closes, which would undo your change.
5. **Change one thing at a time**, then play to check it.

The two kinds of files you will edit

Ends in	What it is	Example
`.tres`	A data card: one fighter, one weapon or one map. Simple "name = value" lines.	`data\weapons\hammer.tres`
`.gd`	Game code. You only touch the lines that start with `const` (a fixed setting) near the top.	`scripts\match\rules.gd`
`.cfg`	Your saved progress (coins, unlocks). Lives outside the game folder.	`progress.cfg`

> **A line that starts with # is just a note.**
> It explains the line under it, and the game ignores it. Those notes are a good way to understand what each setting does.

2. How to open and edit the files

Option A: Notepad (simplest)

1. Open File Explorer and go to the file (for example `D:\gam\data\weapons\hammer.tres`).
2. Right-click it, choose **Open with**, then **Notepad**.
3. Press **Ctrl + F** and type the setting name to jump to it (for example `body_damage`).
4. Change the number, then press **Ctrl + S** to save.

> **Use plain Notepad, not Word.**
> Word adds hidden formatting that breaks the files. Notepad++ or VS Code (both free) are nicer if you want colours and line numbers.

Option B: the Godot editor (best for fighter and weapon cards)

1. Run Godot: `D:\0\Downloads\Godot_v4.7.2-stable_win64.exe\Godot_v4.7.2-stable_win64.exe`
2. Click **Import**, pick `D:\gam\project.godot`, then **Import & Edit**.
3. In the **FileSystem** panel (bottom left), open `data → weapons` or `data → fighters` and double-click a file.
4. Its settings appear in the **Inspector** on the right as boxes. Type new numbers and press **Ctrl + S**.
5. Press **F5** to play the game straight away.

3. Every fighter and their files

Each fighter has a **fighter card** in `data\fighters\` (name, rarity, price, special text) and uses a **weapon card** in `data\weapons\` (damage, speed and how it flies). Damage is shown as **body / headshot**.

In game	Fighter file	Weapon file	Rarity	Damage
Quotuk the Wobbly	quotuk.tres	hammer.tres	Common	30 / 52
Klakla the Lucky	klakla.tres	slingshot.tres	Common	11 / 19 (x3 stones)
Fentuk the Wobbly	fentuk.tres	bow.tres	Common	20 / 35
Bomba the Loud	bomba.tres	bomb.tres	Rare	24 / 38
Pipmo the Mighty	pipmo.tres	spear.tres	Rare	24 / 42
Rango the Swift	rango.tres	boomerang.tres	Rare	22 / 38
Grak the Heavy	grak.tres	axe.tres	Rare	34 / 58
Plopo the Sticky	plopo.tres	plunger.tres	Rare	15 / 26
Megu	megu.tres	nue.tres	Rare	22 / 40
Uryo	uryo.tres	spirit_arrow.tres	Rare	16 / 30
Ryo	ryo.tres	hadoken.tres	Rare	22 / 38
Thundr	thundr.tres	mjolnir.tres	Epic	33 / 55
Capt	capt.tres	star_shield.tres	Epic	26 / 44
Spydr	spydr.tres	web.tres	Epic	16 / 30
Batz	batz.tres	batarang.tres	Epic	24 / 42
Gok	gok.tres	ki_blast.tres	Legendary	20 / 35
Egita	egita.tres	ki_orb.tres	Legendary	19 / 33
Lufi	lufi.tres	meat.tres	Legendary	24 / 42
Ichi	ichi.tres	cleaver.tres	Legendary	32 / 55
Yuj	yuj.tres	iron_ball.tres	Legendary	32 / 52
Goj	goj.tres	blue.tres	Legendary	18 / 32
Akum	akum.tres	beads.tres	Legendary	24 / 42
Jotar	jotar.tres	bearing.tres (Iggi)	Legendary	18 / 34
Dyo	dyo.tres	knife.tres	Legendary	12 / 22 (x2 knives)
Kazma	kazma.tres	kazma_bow.tres	Legendary	14 / 26
Subaro	subaro.tres	minya.tres	Legendary	18 / 32
Aizon	aizon.tres	byakurai.tres (Reiatsu)	Mythic	20 / 35 + small burst

Frierin	frierin.tres	zoltraak.tres	Mythic	14 / 24 orb, 22 / 38 beam

Every fighter starts each match with 100 health, so a 35 headshot takes about a third of a health bar.

4. Damage of basic attacks

Open the fighter's **weapon file** from the table on the previous page. Here is the hammer, with the lines you are most likely to change highlighted:

```
id = &"hammer"
display_name = "Spinning hammer"
max_speed = 1210.0
body_damage = 30
head_damage = 52
count = 1
spread_deg = 0.0
spin = 14.0
sticks = false
...
```

Setting	What it does	Tip
`body_damage`	Damage when it hits the body or legs.	Most weapons are 15 to 34.
`head_damage`	Damage on a headshot.	Usually about 1.7 times the body damage.
`max_speed`	How hard it can be thrown, which sets how far it can reach.	Too low and it can't reach the enemy on big maps. Stay above 1000.
`count`	How many come out per throw (Klakla throws 3).	Each one does full damage, so keep damage low if count is high.
`spread_deg`	Angle between several projectiles.	Only matters when count is more than 1.
`gravity_scale`	How much it drops. 1.0 is normal, 0.5 floats, 0.0 flies dead straight.	Lower makes long shots easier.
`explode_radius`	Size of the blast (bombs only).	Bigger hits from further away.
`sticks`	`true` sticks in the target (arrows, axes), `false` bounces off.	Looks only.

Example: make Grak weaker.

Open `data\weapons\axe.tres` and change `body_damage = 34` to `body_damage = 28` and `head_damage = 58` to `head_damage = 48`. Save.

Special rule for Frierin's dropped beam

Her orb's damage is in `zoltraak.tres`. The beam that drops from it when you press is set in `scripts\match\match_controller.gd`:

```
const ZOLTRAAK_BEAM_DAMAGE := [22, 38]
```

The first number is body damage. The second would be headshot damage, but the beam is set to body hits only.

Bonus damage from levels

Fighters level up when you open a duplicate in a chest. In `match_controller.gd`:

```
const LEVEL_DAMAGE := 0.04   # +4% damage per level
const LEVEL_CHARGE := 0.06   # special charges 6% faster per level
```

5. Damage of specials

Most specials re-use the weapon's damage and multiply it. The multipliers live near the top of `scripts\match\match_controller.gd`, in one long line:

```
const SPECIAL_MULTIPLIERS := {"purple": 1.7, "fuga": 1.8, "gou": 1.7, ...}
```

1.0 means normal damage, **2.0** double, and **0.5** half. Find the word for the special you want and change only its number.

Fighter	Special	Word in the list	Now
Goj	Hollow Purple	"purple"	1.7
Yuj	Fuga	"fuga"	1.8
Akum	Gou Hadoken	"gou"	1.7
Ryo	Shinku Hadoken	"shinku"	1.6
Thundr	God of Thunder	"thunder"	1.5
Capt	Shield Ricochet	"ricochet"	1.3
Spydr	Web Yank	"webyank"	1.4
Batz	Explosive Batarangs (each)	"batbomb"	0.6
Dyo	Knife Storm (each knife)	"knifestorm"	0.35
Megu	Max Elephant	"elephant"	1.5
Uryo	Vollstandig arrows	"voll"	1.35
Subaro	Emilla's icicles (each)	"icefall"	1.0
Frierin	Zoltraak Volley (each sphere)	"zoltraak"	0.4

A few older specials have their own line instead, in the same file:

Fighter	Special	Line
Gok	Mega Wave beam	`const BEAM_MULTIPLIER := 1.5`
Egita	Spirit Orb	`const ORB_MULTIPLIER := 2.4`
Lufi	Gum Barrage (each fist)	`const BARRAGE_MULTIPLIER := 0.4`
Ichi	Moon Fang	`const FANG_MULTIPLIER := 1.7`

> **Specials that hit many times add up fast.**
> Dyo throws 7 knives and Frierin fires 7 spheres, so a small change to their number has a big effect. Change these in small steps, like 0.4 to 0.45.

How fast specials charge

```
const CHARGE_HIT := 0.4          # landing a hit fills 40% of the meter
const CHARGE_HEADSHOT := 0.6     # a headshot fills 60%
const CHARGE_WHEN_HIT := 0.2     # getting hit fills 20%
```

6. Character abilities and passives

All of these are `const` lines in `scripts\match\match_controller.gd`. Press Ctrl + F and search for the name.

Setting	Now	What it controls
`TIME_STOP_THROWS`	2	Jotar: how many throws he gets in stopped time.
`KNIFE_STORM`	7	Dyo: how many knives his special throws.
`BLEED_CHANCE`	0.35	Jotar and Dyo: chance a hit causes bleed (0.35 = 35%).
`BLEED_TURNS` / `BLEED_DAMAGE`	3 / 3	How long bleeding lasts and the damage per turn.
`BURN_TURNS` / `BURN_DAMAGE`	3 / 6	Yuj's Fuga burn.
`LUCKY_THROWS`	2	Kazma: lucky throws from Aqwa's blessing.
`ICEFALL_DROPS`	3	Subaro: icicles per special.
`ICEFALL_RADIUS`	85.0	Subaro: size of the red target circle.
`PRESSURE_HITS`	2	Aizon: Reiatsu hits in a row needed for Spiritual Pressure (the enemy throws their next shot with no aim guide).
`EVOLVED_DEFENCE`	0.8	Aizon: share of damage he takes once evolved (0.8 = 20% less).
`KYOKA_HEAL`	5	Aizon: health he gets back each time Kyoka Suigetsu triggers.
`EVOLVE_HEALTH`	50	Aizon: evolves when he drops to this much health.
`EVOLVED_DAMAGE`	1.3	Aizon: once evolved the enemy's defence is down, and his hits do this many times as much.
`MANA_RELEASE_HEALTH`	50	Frierin: health at which Mana Release (her passives) turns on.
`MANA_BARRIER`	0.75	Frierin: share of damage she takes during Mana Release (0.75 = her barrier blocks a quarter).
`ANALYSIS_STEPS`	4	Frierin: misses needed for Mana Analysis to grow her aim guide to full (it resets after).
`ZOLTRAAK_CIRCLES`	7	Frierin: spheres in her Zoltraak Volley.
`ZOLTRAAK_BURST_RADIUS`	110.0	Frierin: blast size of her drop beam during Mana Release.
`VOLL_THROWS` / `VOLL_HEAL`	3 / 15	Uryo: empowered arrows and heal from Vollstandig.
`HAZARD_EVERY` / `HAZARD_DAMAGE`	5 / 12	A map hazard falls every 5 turns and hits for 12.
`CRATE_CHANCE`	0.4	Chance a power-up crate appears each turn.
`TURN_TIME`	12.0	Seconds per turn in 1v1 friend mode.

> **Example: give Jotar three time-stop throws.**
>
> Find `const TIME_STOP_THROWS := 2` and change it to `const TIME_STOP_THROWS := 3`.

Changing the description to match

If you change how a special works, update its text so players aren't misled. In the fighter's card (for example `data\fighters\jotar.tres`) edit the words inside the quotes on `special_description = "..."`. Keep both quote marks.

7. Health

Everyone's starting health is in `scripts\match\rules.gd`:

```
const MAX_HEALTH := 100
```

Raising it to 150 makes matches longer. Lowering it makes them faster. It applies to every fighter.

8. Coins (gold) and rewards

Give yourself coins right now

1. **Close the game completely.**
2. Press **Windows key + R**, paste this and press Enter:

   ```
   %APPDATA%\Godot\app_userdata\Ink Legends
   ```

3. Open `progress.cfg` with Notepad.
4. Find the line `coins=449` (your number will differ) and change it, for example `coins=99999`.
5. Save, then start the game.

> **Copy progress.cfg before you edit it.**
> If the file gets broken the game may start you from scratch. With a copy you can just put it back.

Earn more coins per match

Open `scripts\match\rewards.gd` and find the function called `coins_for`:

```
if summary["won"]:
    return 50 + DIFFICULTY_BONUS[summary["difficulty"]] + headshot_bonus
return 10 + headshot_bonus
```

The first highlighted number is the reward for a VS AI win, and the second is the reward for a loss. In the same function, the tournament lines use `40` per win, `30` extra per round and `300` for the champion. Each headshot adds 5 coins (`* 5`).

At the top of `rewards.gd` there are more settings:

```
const ACHIEVEMENT_COINS := 50              # per trophy earned
const DAILY_COINS := 150                   # daily challenge reward
const DIFFICULTY_BONUS := [0, 20, 50, 80]    # extra for Easy, Normal, Hard, Expert
```

9. Unlocking fighters

On your current save

1. Close the game and open `progress.cfg` (see section 8).
2. Find the line that starts with `unlocked=[`.
3. Add the fighter's **file name without .tres** in quotes, separated by a comma. For example, to add Frierin:

```
unlocked=["quotuk", "klakla", "frierin"]
```

Every fighter id is in the "Fighter file" column on page 3 (for example `aizon`, `subaro`, `kazma`).

For every new player

Starting fighters are set at the top of `scripts\autoload\save.gd`:

```
const STARTING_FIGHTERS := ["quotuk", "klakla"]
```

Add ids inside the brackets, each in quotes and separated by commas. This only affects brand-new saves, not one you already have.

A fighter's shop price

In the fighter card, change `price = 600`.

10. Rarity tiers and chests

Change a fighter's tier

In the fighter card, change the word inside the quotes on the rarity line. Use exactly one of these five words:

```
rarity = &"legendary"      # common, rare, epic, legendary or mythic
```

Chest prices and odds

In `scripts\match\chests.gd`:

```
"wood":   {"name": "Wooden chest", "price": 150, ..., "odds": [70, 25, 4, 1, 0]},
"silver": {"name": "Silver chest", "price": 400, ..., "odds": [30, 45, 17, 7, 1]},
"gold":   {"name": "Golden chest", "price": 900, ..., "odds": [0, 35, 35, 25, 5]},
```

The five odds numbers are percentages for Common, Rare, Epic, Legendary and Mythic, in that order. **They must add up to 100.**

```
const DUPLICATE_COINS := {...}   # coins for a duplicate of a fully levelled fighter
const LEVEL_UP_COINS := 25       # coins on top of a level-up
const MAX_LEVEL := 5             # highest fighter level
```

11. Maps, wind and gravity

In `scripts\match\rules.gd`:

Setting	Now	What it does
`MAP_SCALE`	1.12	How far apart fighters stand. 1.3 is a bigger map, 1.0 a smaller one.
`GRAVITY`	470.0	How fast throws fall. Higher makes shorter, steeper throws.
`MAX_WIND`	70.0	The strongest wind. 0.0 turns wind off.
`MIN_ANGLE` / `MAX_ANGLE`	-15.0 / 85.0	The lowest and highest angle you can aim.

> **Bigger maps need stronger throws.**
> If you raise `MAP_SCALE` a lot, slow weapons may not reach. Lower `GRAVITY` a little to balance it.

Mountains

In `scripts\match\terrain.gd`:

```
const MOUNTAIN_CHANCE := 0.75   # 75% of matches have a mountain
const MOUNTAIN_MIN := 150.0    # lowest peak
const MOUNTAIN_MAX := 300.0    # highest peak
```

Set `MOUNTAIN_CHANCE` to 0.0 for no mountains, or 1.0 for one every match.

One map's distance

Each map has a card in `data\maps\` (for example `canyon.tres`). Change `min_distance_m` and `max_distance_m`. The game picks a distance between the two for each match.

12. Names, lines and patch notes

To change	File	Line
A fighter's name	data\fighters*name*.tres	`display_name` (long) and `short_name` (short)
Special name and text	same card	`special_name`, `special_description`
Intro lines and taunts	scripts\match\taunts.gd	The lists of sentences in quotes
Fatality captions	scripts\match\fatality.gd	The `NAMES` list near the top
Patch notes	scripts\ui\patch_notes.gd	Add a new block at the top of `NOTES`
Version number	scripts\match\hud.gd	`const VERSION := "v0.2"`

Apostrophes inside text.

Text sits inside double quotes ("..."), so an apostrophe like *I'm* is fine. Never put a double quote inside the text, though: it ends the text early and breaks the file.

13. Testing your change

1. Open the project in Godot (section 2) and press **F5**. The game starts.
2. **Practice** mode is a quick way to try damage and specials, and its **Test Fatality** button plays your fatality straight away.
3. Godot's **Output** and **Debugger** panels at the bottom show any errors in red.

The automatic check-up (optional)

The game has more than 570 automatic checks. To run them, open a Command Prompt (Windows key, type **cmd**) and paste:

```
cd /d D:\gam
"D:\0\Downloads\Godot_v4.7.2-stable_win64.exe\Godot_v4.7.2-stable_win64_console.exe" --
headless --path . -s tests/run_tests.gd
```

It takes a few minutes. The last line should read `... checks, 0 failed`. A few checks expect certain values (for example that every fighter has a known rarity), so a big change can make one fail even when the game still plays fine.

14. When something breaks

What you see	Likely cause	Fix
Game won't start, or Godot shows "Parse Error"	A missing quote, bracket or tab, or a word typed where a number goes	Godot names the file and line. Compare it with your backup and undo the change.
A fighter has vanished	The fighter card is broken	Check the rarity word is one of the five, and every quote is closed.
Coins or unlocks went back	The game was open while you edited the save	Close the game fully, then edit again.
Progress all gone	progress.cfg broke	Copy back your backup of progress.cfg. There is also a `progress.cfg.bak` in the same folder.

Undo with Git (if you use it).

This project is a Git repository. To throw away all changes to one file and go back to the last saved version, run this in the game folder:

```
git checkout -- data/weapons/axe.tres
```

Use your own file path instead of the axe.

15. Putting the game on Android

Godot can build an **.apk** file that installs on any Android phone. You only set it up once, and the game already uses the mobile-friendly renderer and supports touch controls.

One-time setup on your PC

1. **Install Java (OpenJDK 17).** Download "Temurin 17 JDK" for Windows from *adoptium.net* and install it.

2. **Install Android Studio** from *developer.android.com/studio*. Open it once and let it download the Android SDK. Note the SDK folder it shows, usually `C:\Users\you\AppData\Local\Android\Sdk`.

3. **Tell Godot where they are.** In Godot: **Editor → Editor Settings → Export → Android**. Set **Java SDK Path** to the Temurin folder (for example `C:\Program Files\Eclipse Adoptium\jdk-17...`) and **Android SDK Path** to the folder from step 2.

4. **Download export templates. Editor → Manage Export Templates → Download and Install**. They must match your Godot version (4.7.2).

Build the .apk

5. **Project → Export... → Add... → Android.**

6. Fill in **Package → Unique Name** with something like `com.yourname.inklegends`, and set the app name and icon if you like.

7. Leave **Keystore** empty for now. Godot uses its own debug key, which is fine for installing on your own phone.

8. Click **Export Project**, untick **Export With Debug** for a smoother game, and save as `InkLegends.apk`.

Install it on the phone

9. Copy the .apk to your phone over USB, by email, or through Google Drive.

10. Tap it on the phone. Android will ask to allow **Install unknown apps** for the app you opened it with (Files, Drive, etc.). Allow it, then tap **Install**.

> **One-click install while testing.**
> On the phone, turn on **Developer options → USB debugging** (tap *Build number* 7 times in Settings → About phone to show Developer options). Plug it in with a cable, and a small Android button appears at the top right of Godot. Click it to build and launch on the phone straight away.

> **Sharing it with others or the Play Store.**
> For an app other people install, or for Google Play, make your own **release keystore** (Godot's export screen links to the steps) and keep it safe. You need the same key for every future update.

The phone keeps its own save, separate from your PC's. Coins and unlocks don't carry over unless you copy progress.cfg into the app's data folder, which needs a computer and Android's developer tools.

16. iPhone, and playing in a browser

iPhone and iPad

Apple only lets iPhone apps be built on a **Mac**. You need:

- A Mac with **Xcode** (free from the Mac App Store).
- An Apple ID. A free one installs on your own iPhone for 7 days at a time. A paid Apple Developer account ($99/year) is needed for TestFlight or the App Store.

1. Copy the `gam` folder to the Mac and open it in Godot for Mac.
2. Download export templates (same as Android step 4).
3. **Project → Export → Add → iOS.** Enter your **Team ID** (shown in Xcode under your Apple account) and a **Bundle Identifier** like `com.yourname.inklegends`.
4. Export. Godot makes an Xcode project.
5. Open it in Xcode, plug in your iPhone, choose it at the top, and press **Run (▶)**.
6. On the iPhone, trust the developer in **Settings → General → VPN & Device Management**.

The easy way for any phone: play in the browser

A Web build runs in a phone's browser (Chrome or Safari) with nothing to install, and works for iPhone users without a Mac.

1. **Project → Export → Add → Web**, then **Export Project** into an empty folder (for example `D:\gam_web`) as `index.html`.
2. Put everything in that folder into a **.zip**.
3. Make a free account on **itch.io**, choose **Upload new project**, set *Kind of project* to **HTML**, upload the zip and tick **This file will be played in the browser**.
4. Turn on **SharedArrayBuffer support** in the itch.io embed options if Godot's export screen asks for it.
5. Open the page on your phone. You can also keep it private with a password and share the link.

> **Only share builds you're allowed to.**
>
> Some fighters are based on well-known anime and comic characters. That's fine for playing yourself, but publishing the game publicly (Play Store, App Store, a public itch.io page) could bring copyright problems. Keep shared builds private, or swap in your own characters first.

This guide matches Ink Legends v0.2. If settings are renamed in a later version, search the files for the setting's old name or nearby notes.